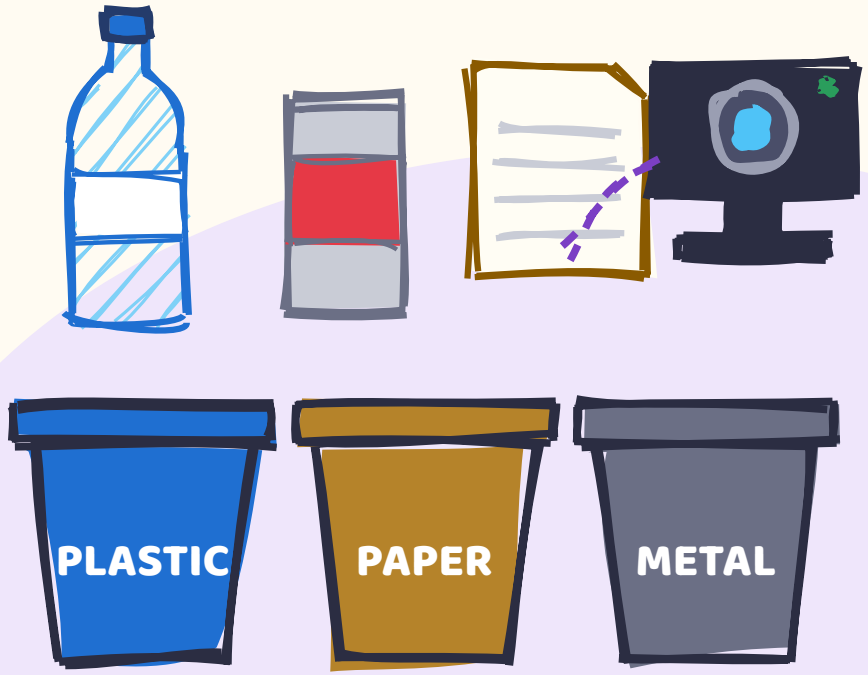
Junior Makers · Class 16

AI Recycling Sorter

Teach a computer to sort our rubbish!

Training data | Teachable Machine | Fair AI



Did you know?
Some recycling plants already use AI cameras and robot arms to pick out plastic, paper and metal.

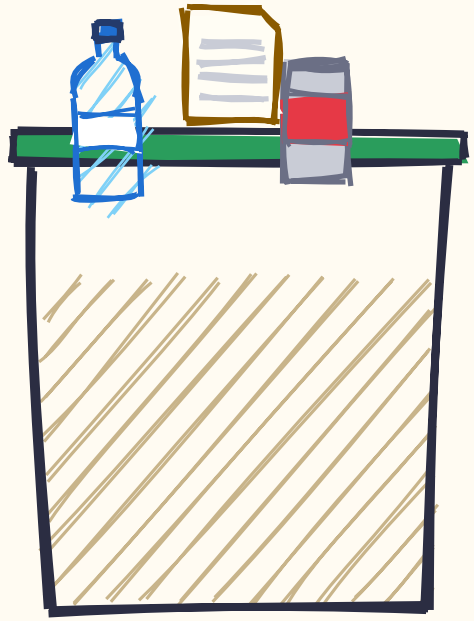
Let's think...

One bin, a hundred kinds of rubbish

The problem

When everything goes into **one bin**, bottles, cans and paper get mixed with food. Mixed rubbish is **hard to recycle**, so a lot of it ends up in landfill or burnt.

Sorting by hand is slow and dirty work.



Sort it all by hand?

Make more bins?

Teach a computer to SEE!

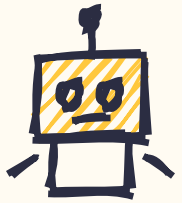
Today: we train our own AI to tell plastic, paper and metal apart, then test how clever (or silly) it is!

Learning aims

Today I will learn to...

1. Explain why **sorting** makes recycling possible
2. Explain how an AI **learns from examples**
3. Collect **training data** and train a model in Teachable Machine
4. **Test** the model and work out its **accuracy**
5. Show how **biased data** makes a **biased AI**, and fix it

Keywords
AI · training data · class · model · accuracy · confidence · bias



Watch and think

Video time!



How AI Works

Code.org · about 2 minutes · youtu.be/Ok-xpKjKp2g

Watch for...

1. How does an AI **learn**?
2. What are **training examples**?
3. Can an AI make mistakes? Why?

Scan to watch

After the video: Our sorter learns plastic, paper and metal. How many examples of each should we show it?

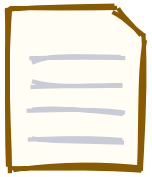
The science

Why sorting matters



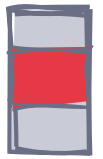
Plastic bottles

Melted into flakes, then made into new bottles, clothes fibres or benches



Paper and card

Soaked into pulp, then pressed into new paper and boxes



Metal cans

Melted and made into new cans, again and again

Each material needs a **different** recycling process. One wrong item can spoil a whole batch, so **sorting comes first**.

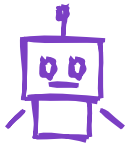
Clean and dry

Food and grease make recycling harder. Rinse it before you bin it!

LEARN

Real world

AI that can see



Sorting robots

AI cameras spot items on a fast belt and robot arms pick them out



Waste pickers

In Pakistan many people sort waste by hand. Smart tools could make the job safer



Phone cameras

Your gallery can search "cat" or "beach" using the same kind of AI



Smart bins

Some new bins look at your rubbish and open the right lid

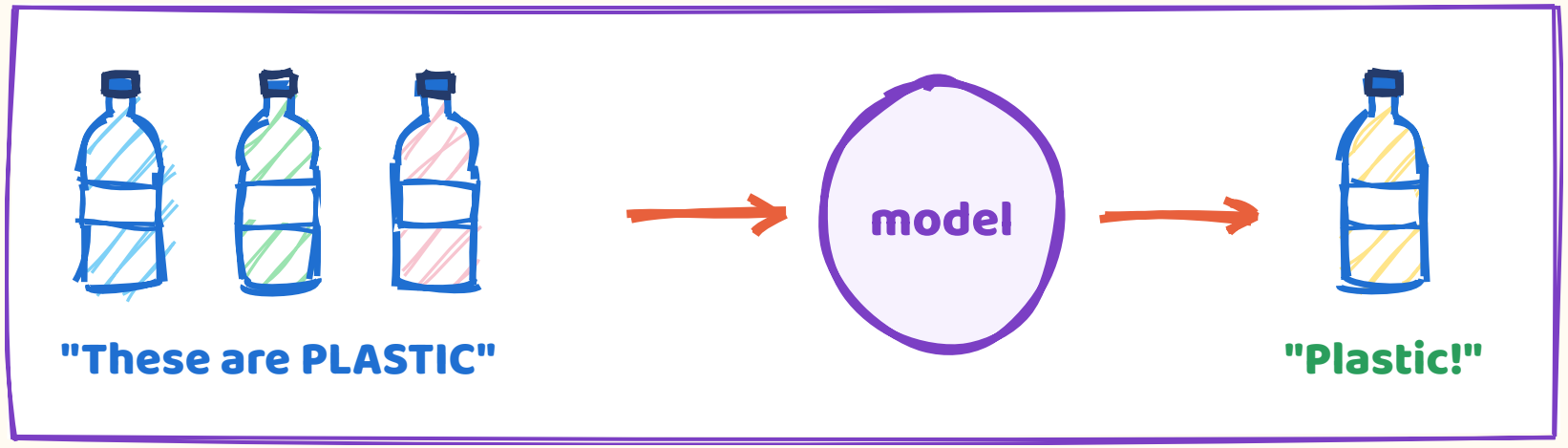
? Where else have you seen a computer **recognise** something from a picture or a camera?

Hint

Face unlock, number-plate cameras, plant ID apps…

How AI learns

Learning from examples



Big idea
An AI is only as good as the examples we give it!

Like teaching a little brother

You don't give him rules. You show him **lots of examples**: "this is a cat, this is a cat, this is a dog". Soon he spots cats he has **never seen**.

The words

Class: a group (Plastic)
Training data: the example photos
Model: what the AI learnt

Be a data scientist

How accurate is our AI?

Item	Real answer	AI said	Correct?
New plastic bottle	_____	_____	yes / no
Crushed can	_____	_____	yes / no
Cereal box	_____	_____	yes / no
Yogurt cup	_____	_____	yes / no
Foil tray	_____	_____	yes / no
Newspaper	_____	_____	yes / no

Accuracy

correct ÷ total × 100

5 ÷ 6 × 100 = 83%

Compare

Which team got the highest accuracy? What did they do differently?

Fill in your **printed results sheet**. We will use it again after we improve the AI.

Think harder

Items that confuse humans AND AI

?

Crisp packet

plastic AND a thin metal layer inside

?

Juice carton

paper, plastic and foil stuck together

?

Greasy pizza box

paper, but the grease can spoil the batch

Design a safety rule

If the AI's confidence is **below 70%**, it should say **"Ask a human"** instead of guessing. Good AI knows when it is **not sure**.

Discuss

Should an AI ever make a decision with no human checking?

Be a good digital citizen

Using cameras and AI responsibly

Ask first: never photograph a friend's face without asking.

Check the camera: point it at the items, not at people or private papers.

Keep it local: do not upload or share models with photos of people in them.

Be honest: tell people when an AI made a decision, and how accurate it is.



Remember

AI is a tool. People are responsible for how it is used.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Teachable Machine
2. Collect your training data
3. Be the AI!
4. Train the model
5. Test it with NEW items
6. When AI learns the wrong lesson
7. Make your AI fairer and smarter

Our tool

Teachable Machine

A free website from Google: **teachablemachine.withgoogle.com**. Choose **Image Project**, then **Standard image model**.



1. Gather

Take photos of each class with the webcam



2. Train

The computer studies the photos



3. Test

Show it NEW items and see what it says



The training happens **inside your browser**. No coding needed and no account for today.

Teacher check

Allow the webcam pop-up. Test the site on school Wi-Fi before class.

Step 1

Collect your training data

Class: Plastic

bottles, cups, food tubs

Class: Paper

newspaper, card, boxes

Class: Metal

cans, foil trays, tins

Class: Nothing

empty table, your hand

Good photo tips

About **50 photos** per class

Turn and tilt each item

Use **many different** items

Hold the **Record** button and move slowly

Why a Nothing class?

Without it, the AI **must** pick plastic, paper or metal, even when the table is empty!

Team jobs

Holder · Recorder · Checker. Swap jobs each class!

Class game

Be the AI!

How to play

1. One volunteer is the "AI" and waits outside.
2. The class picks a **secret rule** (e.g. "metal things").
3. The AI comes back. We show items and say only **"yes"** or **"no"**.
4. After 6 examples, the AI must guess new items.
5. Can the AI guess the secret rule?

Then discuss

Did the AI learn the **right** rule, or a wrong one like "shiny things"? Would **more** examples help? Which ones?



Big idea

That is exactly how a computer learns: examples and feedback!

Step 2

Train the model



It usually takes **under a minute**. Then the **Preview** panel switches on with your webcam.

Press Train Model

The computer looks at **every photo many times**. Each time it checks its guesses and adjusts itself a little. Each full pass is called an **epoch**.

Do not switch tabs!

Keep the Teachable Machine tab open while it trains, or it may pause.

While you wait

Predict: which class will your AI find hardest? Write it down!

Step 3

Test it with NEW items

Preview panel



Golden rule of testing

Test with items the AI has **never seen**. Testing with training items is like giving a student the exam answers before the exam!



Confidence

The bars show how **sure** the AI is. 86% plastic = "I am fairly sure this is plastic".

Try to trick it

Crushed can · bottle with no label · coloured card · item far away · dim light

Experiment

When AI learns the wrong lesson



trained: only CLEAR bottles



green bottle?

"Metal?!"

Biased data
makes a biased AI. Always ask: who or what is MISSING from my examples?

What happened?

The AI never saw a coloured bottle. It may have learnt "plastic = **see-through**" instead of "plastic = bottle shape". This is called **bias**.

Why it matters

Real AIs trained on unfair data can work **worse for some people**, for example face unlock that fails for some faces.

Fix it

Make your AI fairer and smarter

- ☑ **Variety:** different colours, sizes, brands and shapes
- ☑ **Balance:** about the same number of photos in every class
- ☑ **Real life:** crushed, dirty, half-hidden, near and far
- ☑ **Lighting:** bright, dim, and by the window
- ☑ **Background:** different tables and hands, not always the same

Then…

Add the new photos, **train again**, and test with the **same** items. Did accuracy go up?

Engineers say

Better data beats a cleverer computer, most of the time!

Level up!

AI sorter challenges

Choose your challenge



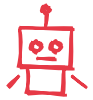
New class: add Glass or Food waste. Does accuracy drop?



Beat your score: get above 90% accuracy on the 6 test items.



Swap and test: test another team's model with YOUR tricky items.



Make it move: export the model and connect it to a servo that points at the right bin (Young Innovators level!).

Think like an engineer

A real plant sorts **thousands** of items an hour. What would go wrong if the AI was only 80% accurate?

Poster task

Make a poster for your school: which bin does each item go in?

Show what you know

What did we learn?

- [x] Sorting makes recycling possible
- [x] AI learns from examples called training data
- [x] Test with NEW items to find the real accuracy
- [x] Biased data makes a biased AI
- [x] Good AI says "ask a human" when it is not sure

Exit ticket: name ONE way to make your AI fairer.

At home

Sort one day of your family's rubbish into plastic, paper, metal and food. Which pile was biggest?



Batch complete!
Get ready to show your best project.